# LAB SHEET -3

***1. Create an HTML page using JavaScript to accept two numbers and perform addition, subtraction, multiplication, and division.***

JPG

```
<script>
    function calculate() {
        let n1 = parseFloat(document.getElementById("num1").value);
        let n2 = parseFloat(document.getElementById("num2").value);

        let add = n1 + n2;
        let sub = n1 - n2;
        let mul = n1 * n2;
        let div = n2 !== 0 ? (n1 / n2) : "Cannot divide by zero";

        document.getElementById("result").innerHTML =
            "<br>Addition: " + add +
            "<br>Subtraction: " + sub +
            "<br>Multiplication: " + mul +
            "<br>Division: " + div;
    }
</script>
```

***2. Write a JavaScript program to accept a student's marks and display the grade according to the following criteria: 90–100: A+, 80–89: A, 70–79: B, 60–69: C, Below 60: Fail.***

JPG

```
<script>
    function checkGrade() {
```

```
    let marks = parseFloat(document.getElementById("marks").value);
    let grade = "";

    if (marks >= 90 && marks <= 100) {
        grade = "A+";
    } else if (marks >= 80 && marks < 90) {
        grade = "A";
    } else if (marks >= 70 && marks < 80) {
        grade = "B";
    } else if (marks >= 60 && marks < 70) {
        grade = "C";
    } else if (marks < 60 && marks >= 0) {
        grade = "Fail";
    } else {
        grade = "Invalid Marks";
    }

    document.getElementById("gradeResult").innerText = "Grade: " + grade;
  }
</script>
```

***3. Write a JavaScript program to accept a number from the user and display its multiplication table using a loop.***

JPG

```
<script>
  function generateTable() {
    let n = parseInt(document.getElementById("tabNum").value);
    let output = "";

    for (let i = 1; i <= 10; i++) {
```

```
        output += n + " x " + i + " = " + (n * i) + "<br>";
    }

    document.getElementById("tableResult").innerHTML = "<br>" + output;
}
</script>
```

***4. Create a JavaScript function to calculate the factorial of a number entered by the user.***

JPG

```
<script>
    function getFactorial(n) {
        let fact = 1;
        for (let i = 1; i <= n; i++) {
            fact *= i;
        }
        return fact;
    }

    function findFactorial() {
        let num = parseInt(document.getElementById("factNum").value);
        if (num < 0) {
            document.getElementById("factResult").innerText = "Factorial not defined for negative numbers";
        } else {
            let ans = getFactorial(num);
            document.getElementById("factResult").innerText = "Factorial of " + num + " is: " + ans;
        }
    }
```

```
</script>
```

***5. Create a simple Student Registration Form and use JavaScript to validate Name, Email, Mobile Number, and Password before submitting the form.***

JPG

```
<script>
   function validateForm() {
      let name = document.getElementById("name").value.trim();
      let email = document.getElementById("email").value.trim();
      let mobile = document.getElementById("mobile").value.trim();
      let password = document.getElementById("password").value.trim();
      let msg = document.getElementById("errorMsg");

      if (name === "") {
         msg.innerText = "Name cannot be empty";
         return false;
      }

      let emailPattern = /^[^ ]+@[^ ]+\.[a-z]{2,3}$/;
      if (!email.match(emailPattern)) {
         msg.innerText = "Please enter a valid email address";
         return false;
      }

      let mobilePattern = /^[0-9]{10}$/;
      if (!mobile.match(mobilePattern)) {
         msg.innerText = "Mobile number must be exactly 10 digits";
         return false;
      }
```

```
        if (password.length < 6) {
            msg.innerText = "Password must be at least 6 characters long";
            return false;
        }

        alert("Registration Successful!");
        return true;
    }
</script>
```